Problem Set 5E

Distances, angles and intersections

Do these after the slides for Section 5E. Textbook Exercise 5E. Answers are on the last page. Show full working in your exercise book.

Skills

1 Find the distance from the point $(2, 1, -3)$ to the plane $2x + 2y - z = 5$.

2 Find the distance between the parallel planes $x - 2y + 2z = 1$ and $x - 2y + 2z = 10$.

3 The line $\mathbf{r} = 2\mathbf{i} - \mathbf{j} + t(\mathbf{i} + \mathbf{j} - 2\mathbf{k})$ meets the plane $x + 2y - z = 10$. Find:

(a) the point of intersection

(b) the angle between the line and the plane, correct to one decimal place

4 Find, correct to one decimal place, the acute angle between the planes $2x - y + 2z = 4$ and $x + 2y - 2z = 1$.

5 Find a vector equation of the line of intersection of the planes $x + y + z = 6$ and $x - y + 2z = 5$.

Exam style

6 Find the shortest distance between the skew lines $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + \lambda(\mathbf{i} + \mathbf{k})$ and $\mathbf{r} = 3\mathbf{k} + \mu(\mathbf{j} + \mathbf{k})$.

7 The point $A(3, 1, 4)$ is reflected in the plane $x - y + 2z = 4$. Find the foot of the perpendicular from A to the plane, the distance from A to the plane, and the image of A.

8 Show that the line $\mathbf{r} = \mathbf{i} + \mathbf{k} + t(\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})$ is parallel to the plane $2x + y - 2z = 6$, and find the distance between them.

Problem Set 5E: answers

1 $\frac{4}{3}$

2 3

3 (a) $(4, 1, -4)$ (b) $\sin\varphi = \frac{5}{6}$, $\varphi = 56.4°$

4 $\cos\theta = \frac{4}{9}$, $\theta = 63.6°$

5 $\mathbf{r} = 7\mathbf{i} - \mathbf{k} + t(3\mathbf{i} - \mathbf{j} - 2\mathbf{k})$ (other forms are possible)

6 $\mathbf{n} = -\mathbf{i} - \mathbf{j} + \mathbf{k}$; distance $= \dfrac{6}{\sqrt{3}} = 2\sqrt{3}$

7 Foot $(2, 2, 2)$; distance $\sqrt{6}$; image $(1, 3, 0)$

8 $\mathbf{d} \cdot \mathbf{n} = 2 + 2 - 4 = 0$; distance $= \dfrac{|2 - 2 - 6|}{3} = 2$